电磁波在等离子体中传播.

1) $\vec{B}=0$. $KT=0$. 平面波.

$$\omega_p^2 = \frac{n_0 e^2}{m\varepsilon_0}.$$

KT_e: $\omega^2 = \omega_p^2 + \frac{3}{2}mv_{th}^2$

KT_i: $\omega^2 = k^2\left(\frac{KT_e + \gamma KT_i}{M}\right)$

$\vec{B}_0$: $\omega^2 = \omega_p^2 + \omega_c^2$.

$$\Omega^2 = \omega_c \Omega_c.$$

□ $\vec{B}_0 = 0$ $\vec{k} \perp \vec{E}_1 \Rightarrow \vec{B}_1$.

电磁波.

真空情形, $\rho = 0$ $\vec{j} = 0$.

$$\nabla \times \vec{E} = \frac{\partial \vec{B}}{\partial t} \qquad \nabla \cdot \vec{E} = 0. \quad \nabla \cdot \vec{B} = 0.$$

$$\nabla \times \vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{j}$$

$\vec{E} = \vec{E}_0 + \vec{E}_1 = \vec{E}_1 \qquad \vec{B} = \vec{B}_1$

$\therefore \qquad \nabla \times \vec{E}_1 = \frac{\partial \vec{B}_1}{\partial t} \,, \qquad \nabla \times \vec{B}_1 = \frac{1}{c^2} \frac{\partial \vec{E}_1}{\partial t}$

$$\nabla \times (\nabla \times \vec{B}_1) = -\frac{1}{c^2} \frac{\partial}{\partial t} \vec{B}_1$$

$$-\vec{k} \times (\vec{k} \times \vec{B}_1) = +\frac{\omega^2}{c^2} \cdot \vec{B}_1$$

由 $a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$.

$\therefore \qquad k^2 \vec{B}_1 = \frac{\omega^2}{c^2} \vec{B}_1 \,, \quad c^2 k^2 = \omega^2$

$$\nabla \times \vec{E}_1 = -\frac{\partial}{\partial t} \vec{B}_1 \,, \quad i\vec{k} \times \vec{E}_1 = +i\omega \vec{B}_1 \,.$$

$$|k| |\vec{E}_1| = \omega |\vec{B}| \,, \quad c = \frac{\omega}{k} = \frac{|E_1|}{|B_1|}$$

· $E_1 \gg B_1$, 仅考虑 E_1.

等离子体中，

- 电子位移产生电流.

$$\nabla \times \vec{B}_1 = \frac{1}{c^2}\frac{\partial \vec{E}_1}{\partial t} + \mu_0 \vec{j}_1$$

$$\Rightarrow \qquad \nabla \times \left(\frac{\partial}{\partial t}\vec{B}_1\right) = \frac{1}{c^2}\frac{\partial^2 \vec{E}_1}{\partial t^2} + \mu_0 \frac{\partial}{\partial t}\vec{j}_1 .$$

$$\nabla \times (\nabla \times \vec{E}_1) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}_1) = -\frac{\partial}{\partial t}\left(\frac{1}{c^2}\frac{\partial \vec{E}_1}{\partial t} + \mu_1 \vec{j}\right).$$

$$\vec{k} \times (\vec{k} \times \vec{E}_1) = \frac{\omega^2}{c^2}\vec{E}_1 + i\omega\mu_0 \vec{j}_1$$

运动方程：

$$m\frac{d\vec{v}_{e1}}{dt} = -e\vec{E}_1$$

$$\text{left} = m\left(\frac{\partial \vec{v}_{e1}}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial \vec{v}_{e1}}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial \vec{v}_{e1}}{\partial z}\frac{\partial z}{\partial t} + \frac{\partial \vec{v}_{e1}}{\partial t}\right) \quad \text{全微分}$$

$$= m\left(\frac{\partial \vec{v}_{e1}}{\partial x}\cdot v_{e1x} + \frac{\partial \vec{v}_{e1}}{\partial y}\cdot v_{e1y} + \frac{\partial \vec{v}_{e1}}{\partial z}\cdot v_{e1z} + \frac{\partial \vec{v}_{e1}}{\partial t}\right)$$

$$= m\frac{\partial \vec{v}_{e1}}{\partial t}$$

$$\therefore \qquad i\omega m\vec{v}_{e1} = e\vec{E}_1, \quad \vec{j} = -en_0\vec{v}_{e1} = -\frac{e^2 n_0 \vec{E}_1}{im\omega} = \frac{in_0 e^2}{m\omega}\vec{E}_1 .$$

$$-\vec{k}(\vec{k}\cdot\vec{E}_1) - k^2\vec{E}_1 = \frac{\omega^2}{c^2}\vec{E}_1 + \left(\frac{-\omega\mu_0 n_0}{m\omega}e^2\vec{E}_1\right).$$

$$k^2\vec{E}_1 = \frac{\omega^2}{c^2}\vec{E}_1 - \frac{\mu_0 n_0 e^2}{m}\vec{E}_1 = \frac{\omega^2}{c^2}\vec{E}_1 - \frac{n_0 e^2}{c^2\varepsilon_0 m}\vec{E}_1$$

$$= \frac{\omega^2}{c^2}\vec{E}_1 - \frac{\omega_p^2}{c^2}\vec{E}_1$$

$\therefore$ $$\omega^2 = \omega_p^2 + c^2k^2$$

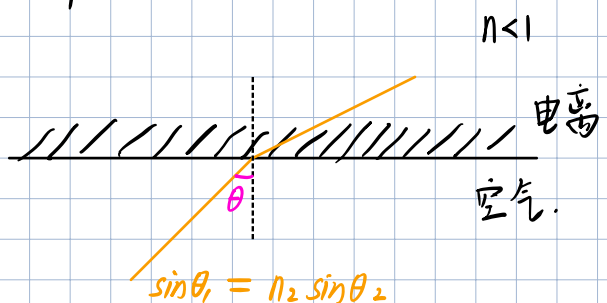
phase velocity: $$v_p = \frac{\omega}{k} = \frac{\sqrt{\omega_p^2 + c^2k^2}}{k} = \sqrt{c^2 + \frac{\omega_p^2}{k^2}} > c.$$

group velocity: $$v_g = \frac{d\omega}{dk} = \frac{c^2k}{\omega} = \frac{c^2}{v_p} < c.$$

$$n = \frac{c}{v_p} = \frac{ck}{\omega} \Rightarrow n^2 = \frac{c^2k^2}{\omega^2} = \frac{\omega^2 - \omega_p^2}{\omega^2}$$

$$\Rightarrow \quad n = \sqrt{1 - \omega_p^2/\omega^2}$$

考虑一定 ω, $\omega > \omega_p$. $n > 1$.



$\sin\theta_1 = n_2 \sin\theta_2$

$\sin\theta_c = n_2$, $\theta > \theta_c$ 全反射

无线电波工作原理.

考虑 $k = \omega\sqrt{\varepsilon_0\mu_0}$.

$$\nabla \times \vec{B} = \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t} + \mu_0\vec{j}$$
$$= -\mu_0\varepsilon_0 i\omega\vec{E} + \mu_0\sigma\vec{E}$$
$$= -i\omega\mu_0\left(\varepsilon_0\vec{E} + \frac{i\sigma}{\omega}\vec{E}\right).$$

$$= -i\omega\mu_0 \underbrace{(\varepsilon_0 + \frac{i\sigma}{\omega})}_{\varepsilon} \vec{E}$$

$$k = \omega\sqrt{\mu_0\varepsilon} = \frac{n\omega}{c}$$

$$\Rightarrow \quad \mu_0\varepsilon = \frac{n^2}{c^2} = \varepsilon_0\mu_0 n^2$$

$$\Rightarrow \quad \varepsilon = \varepsilon_0 n^2$$

$$\varepsilon_r =$$

考虑 $\omega = \omega_p$. $n_0 = \frac{\omega^2 \varepsilon_0 m}{e^2}$ 无法传播. ω_p 为截止频率.

可通过测量截止频率测等离子体密度.

$\omega < \omega_p$. 趋肤效应.

$\omega^2 - \omega_p^2 = c^2k^2 < 0$. k 为纯虚数.

$k = i\alpha$. $\alpha \in R$

$\exp(ikx - \omega t) = \underbrace{e^{-\alpha x}}_{\text{指数衰减}} e^{-i\omega t}$. 趋肤效应.

